

Analyzing Fashion Trends Across Market Segments: A Bayesian Approach

Isaiah Cabrinha

Dr. Garritt Page

Introduction

In this report, we are interested in analyzing trends in fashion styles over time. The main purpose of this research is to examine how multiple clothing companies compare in the distribution of their products across the various stages of fashion trends. We then group each company into one of three categories—high-end, middle, and lower-end—based on its overall market positioning, including factors such as price, reputation, and fashion appeal. By comparing these groups, we aim to explore how a company's market segment may influence the distribution of its products across the different trend cycles.

Exploratory Data Analysis

We've taken a dataset from Kaggle that analyzes the fashion trends of winter clothing from the winter season of 2023, 2024, and 2025. The dataset also has around 150 observations, which doesn't provide a substantial amount of evidence towards our question. The primary use of this dataset was to formulate our research question and begin a foundation for what we would like to explore concerning fashion trends over time. The following exploratory data analysis figures were produced:

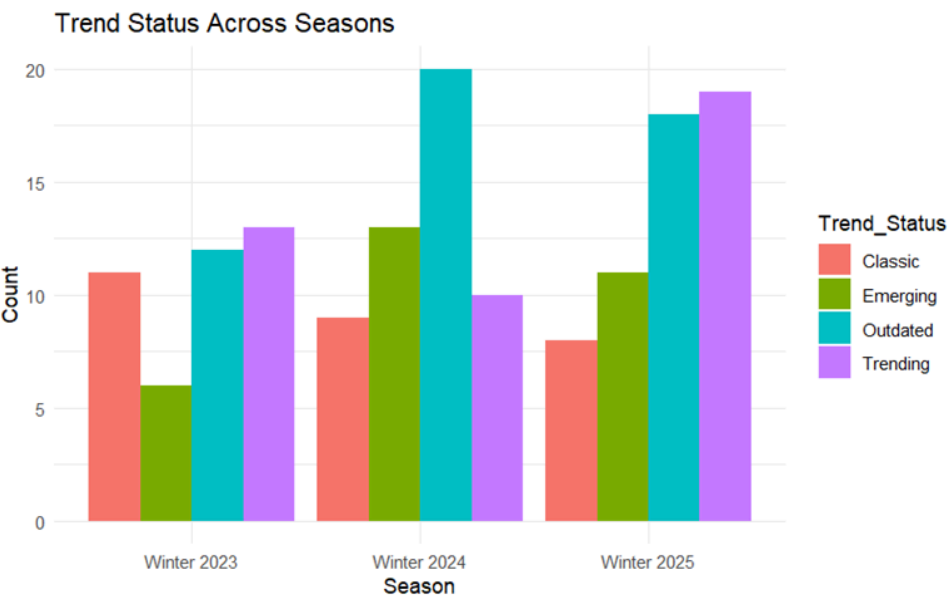


Figure 1: Displays the Trend status of Classic, Emerging, Outdated, and Trending counts over the three winter seasons.

This figure gives us a great gauge into how the trending statuses are constantly changing and evolving over each year. It seems to produce a sort of cycle that we are interested in trying to predict and model how to optimize the most productive and efficient cycle.

	Classic	Emerging	Outdated	Trending
Adidas	1	3	2	4
Gucci	3	2	6	3
H&M	0	2	4	9
Levi's	3	0	3	2
Mango	7	4	11	3
Nike	2	3	4	4
North Face	5	7	6	2
Prada	2	6	4	4
Uniqlo	4	0	6	6
Zara	1	3	4	5

Figure 2: Displays the relationship between brands and each category of trending status

The above table provided a good foundation for fitting our model. For example, we observed that the brand Mango had the highest number of outdated clothing items, suggesting that it did not balance the different stages of fashion trends as well as the other brands. In contrast, most of the remaining brands exhibited relatively balanced counts across the various trend categories.

Beta-Binomial Model

To further investigate changes in fashion trends over time, we constructed a Beta-Binomial model in which a "success" was defined as a clothing item being classified as "Classic." A Beta(5,5) prior was selected to reflect a weak prior belief that approximately half of winter fashion items would be classified as Classic while still allowing the observed data to have the greatest influence on the posterior distribution.

Season was treated as the grouping variable in the analysis. Rather than estimating a single probability that an item was classified as Classic across the entire dataset, separate Beta-Binomial models were fit for Winter 2023, Winter 2024, and Winter 2025. For each season, the observed number of Classic items was combined with the Beta(5,5) prior to produce a posterior Beta distribution. These posterior distributions were then evaluated over a sequence of probability values between 0 and 1 and plotted together. Comparing the posterior distributions allowed us to assess how the probability of an item being classified as Classic changed across the three winter seasons while accounting for the uncertainty in each estimate.

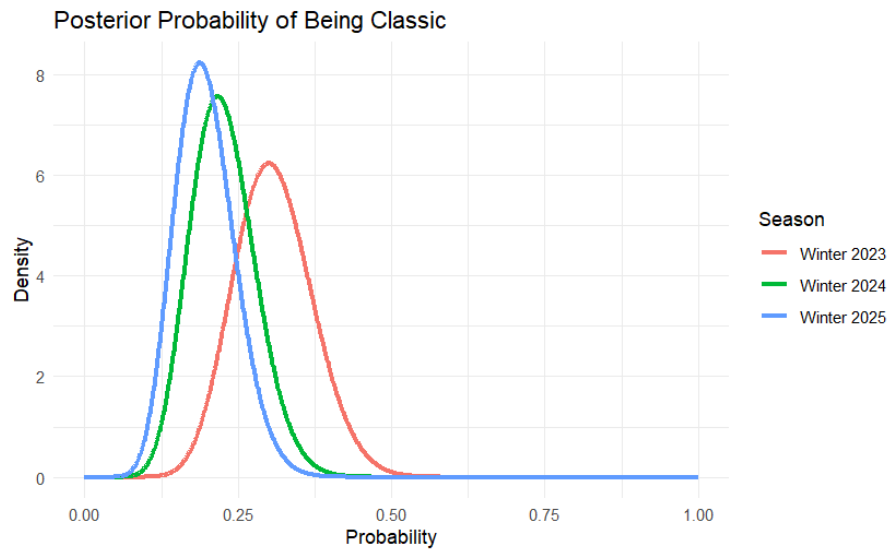


Figure 3: Posterior distributions for the probability that a winter fashion item is classified as “Classic” for Winter 2023, Winter 2024, and Winter 2025.

Figure 3 shows the posterior distributions shift progressively to the left from Winter 2023 to Winter 2025, indicating a decreasing probability that an item is classified as Classic over time. Although the distributions overlap, suggesting some uncertainty in the estimates, the posterior means indicate a downward trend in the prevalence of Classic items across the three winter seasons.

Market Segment Classification

To further investigate whether market positioning is associated with different fashion trend cycles, we grouped the brands into three categories based on their general price point, reputation, and fashion positioning: high-end, middle, and lower-end. The high-end group consisted of Gucci and Prada, the middle group consisted of Adidas, Nike, Levi's, and North Face, and the lower-end group consisted of H&M, Zara, Mango, and Uniqlo. These groupings were based on how each brand typically describes and markets its products, as well as its general price point and reputation within the fashion industry. The high-end group primarily consists of luxury brands that emphasize premium materials, craftsmanship, and exclusive products. The middle group markets its products as higher-quality or premium compared with lower-end brands, while generally remaining more affordable than luxury brands. The lower-end group primarily consists of fast-fashion and value-oriented brands that focus on affordability, accessibility, and producing a wide variety of styles at a relatively rapid rate in order to appeal to a broader audience. These classifications are intended to represent general market positioning rather than precise price classifications, as individual products within each brand may vary considerably in price and style.

Multinomial-Dirichlet Model

For each market segment, the number of clothing items belonging to each trend category was combined across the brands within that segment. Because each clothing item falls into one of

four possible trend categories—Classic, Emerging, Outdated, or Trending—we modeled the resulting counts using a multinomial distribution. Let $M = 4$ represent the four trend categories and let N represent the total number of clothing items within a market segment. The vector of observed counts was modeled as a multinomial random variable with probability vector π , where each element of π represents the probability that an item from that market segment belongs to a particular trend category. The resulting table is shown below.

	Classic	Emerging	Outdated	Trending
High-End	5	8	10	7
Lower-End	12	9	25	23
Middle	11	13	15	12

Figure 4: The total combined counts of all the brands in their respective groups

A Dirichlet(1,1,1,1) prior was selected for the probability vector. This prior gives equal prior weight to all possible combinations of probabilities across the four trend categories, providing a relatively noninformative prior that allows the observed data to have the greatest influence on the posterior distribution. The resulting posterior distribution was also Dirichlet, with each prior parameter increased by the corresponding observed category count.

From each posterior distribution, we calculated the posterior mean for each trend category. These values represent our estimated probabilities that a randomly selected clothing item from a particular market segment belongs to each of the four trend categories after incorporating both the prior information and the observed data.

	Classic	Emerging	Outdated	Trending
High-End	0.176	0.265	0.324	0.235
Lower-End	0.178	0.137	0.356	0.329
Middle	0.218	0.255	0.291	0.236

Figure 5: The posterior mean probabilities of each trend status within each market segment.

Interestingly, Outdated has the highest posterior probability within each of the three groups, with probabilities of 0.324 for high-end, 0.356 for lower-end, and 0.291 for middle brands. The lower-end group also has a relatively high probability of being classified as Trending (0.329), while the middle group has the highest probabilities for Classic (0.218) and Emerging (0.255) compared with the other groups. The probability of an item being classified as Classic is relatively similar across all three groups, ranging from 0.176 to 0.218. These differences suggest that market positioning may be associated with how clothing items are distributed across the different stages of fashion trends.

Monte Carlo Posterior Comparisons

To further compare the posterior probabilities between market segments, we used a Monte Carlo simulation. For each market segment, 10,000 samples were generated from its posterior Dirichlet distribution. The simulated probabilities were then used to calculate the posterior distribution of the difference between each pair of market segments for each trend category. This allowed us to estimate the posterior mean difference, 95% credible interval, and probability that one market segment had a greater probability than another. The results of this simulation are shown below.

Posterior Comparisons Between Market Segments				
Differences in probability across trend categories				
Trend	Comparison	Mean Difference	95% Credible Interval	P(First > Second)
Classic	High-End - Middle	-0.042	(-0.202, 0.131)	29.3%
Classic	High-End - Lower-End	-0.001	(-0.145, 0.159)	47.5%
Classic	Middle - Lower-End	0.041	(-0.096, 0.183)	71.7%
Emerging	High-End - Middle	0.010	(-0.171, 0.199)	53.8%
Emerging	High-End - Lower-End	0.127	(-0.034, 0.300)	93.7%
Emerging	Middle - Lower-End	0.117	(-0.016, 0.257)	95.6%
Outdated	High-End - Middle	0.033	(-0.163, 0.233)	62.6%
Outdated	High-End - Lower-End	-0.033	(-0.220, 0.164)	35.8%
Outdated	Middle - Lower-End	-0.066	(-0.224, 0.098)	20.8%
Trending	High-End - Middle	-0.001	(-0.176, 0.188)	48.5%
Trending	High-End - Lower-End	-0.093	(-0.262, 0.091)	15.4%
Trending	Middle - Lower-End	-0.092	(-0.246, 0.065)	12.3%

Figure 6: Table of the mean posterior differences, 95% credible interval and probability comparisons between each market segment for each trend status

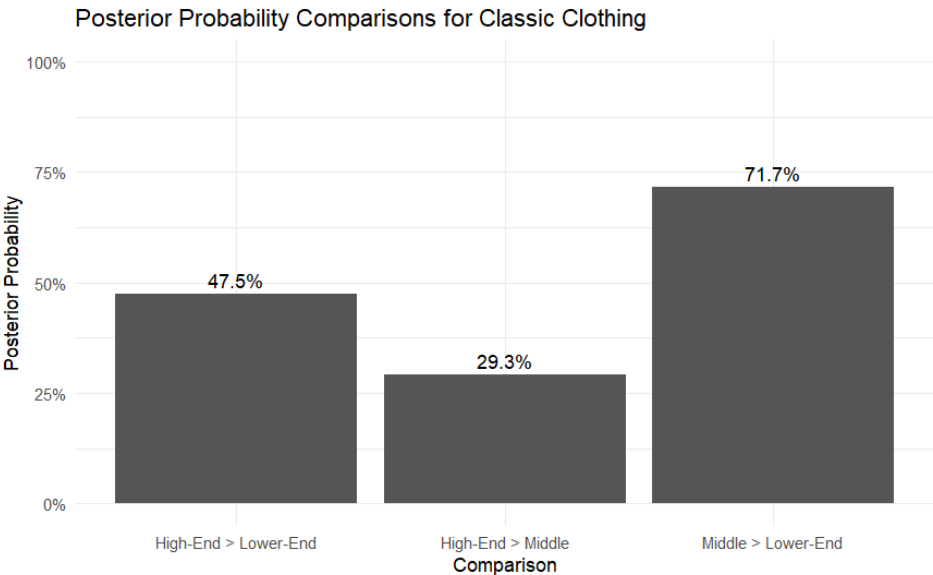


Figure 7: Bar chart of the posterior probability comparisons for the classic trend status

For the Classic category, the difference between High-End and Middle brands had a posterior mean of -0.042, with a 95% credible interval of (-0.202, 0.131) and a posterior probability of 29.3% that High-End brands had a higher probability of being Classic. The difference between High-End and Lower-End brands was -0.001, with a 95% credible interval of (-0.145, 0.159) and a posterior probability of 47.5% that High-End brands had a higher probability. Finally, the difference between Middle and Lower-End brands was 0.041, with a 95% credible interval of (-0.096, 0.183) and a posterior probability of 71.7% that Middle brands had a higher probability. Overall, these results provide relatively little evidence of a substantial difference in the probability of an item being classified as Classic between the three market segments.

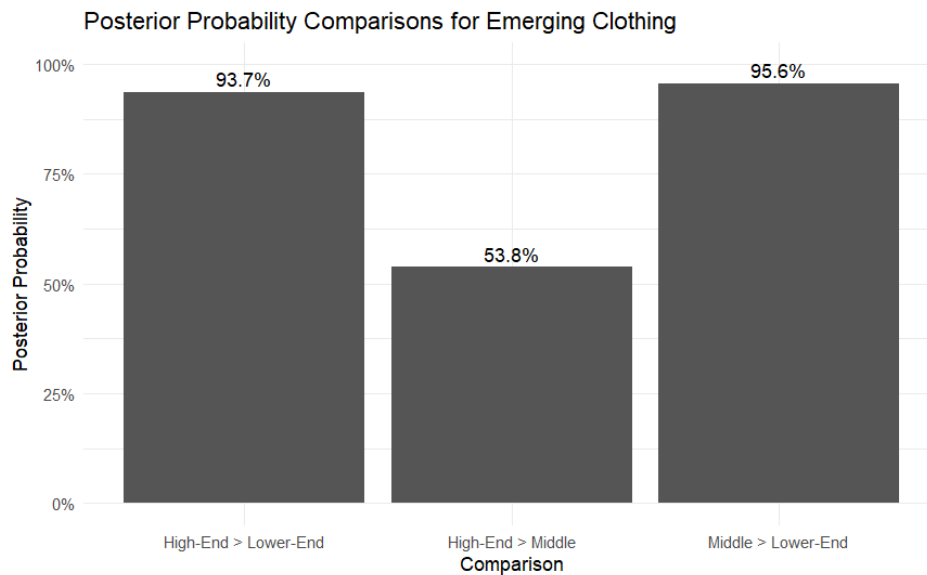


Figure 8: Bar chart of the posterior probability comparisons for the emerging trend status

For the Emerging category, the comparisons showed stronger differences between the groups. The difference between High-End and Middle brands had a posterior mean of 0.010, with a 95% credible interval of (-0.171, 0.199) and a posterior probability of 53.8% that High-End brands had a higher probability. In comparison, High-End brands had an estimated 0.127 higher probability of being classified as Emerging than Lower-End brands, with a 95% credible interval of (-0.034, 0.300) and a posterior probability of 93.7% that the High-End probability was greater. Similarly, the difference between Middle and Lower-End brands was 0.117, with a 95% credible interval of (-0.016, 0.257) and a posterior probability of 95.6% that Middle brands had a higher probability of being classified as Emerging. These results suggest relatively strong posterior evidence that both High-End and Middle brands may have a greater proportion of Emerging items than Lower-End brands.

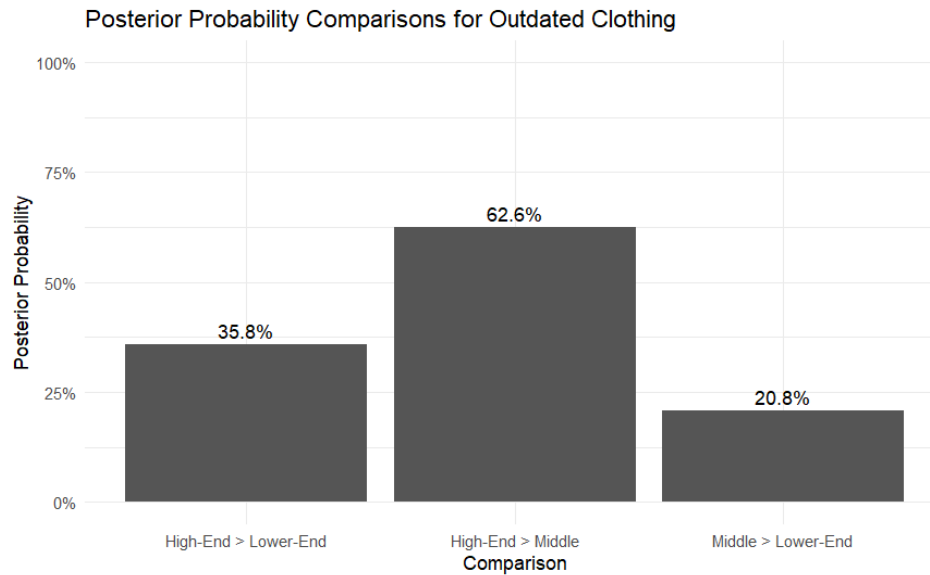


Figure 9: Bar chart of the posterior probability comparisons for the outdated trend status

For the Outdated category, the difference between High-End and Middle brands had a posterior mean of 0.033, with a 95% credible interval of (-0.163, 0.233) and a posterior probability of 62.6% that High-End brands had a higher probability. The comparison between High-End and Lower-End brands had a posterior mean of -0.033, with a 95% credible interval of (-0.220, 0.164) and a posterior probability of 35.8% that High-End brands had a higher probability. This corresponds to a posterior probability of approximately 64.2% that Lower-End brands had a higher probability of being classified as Outdated than High-End brands. The difference between Middle and Lower-End brands was -0.066, with a 95% credible interval of (-0.224, 0.098) and a posterior probability of 20.8% that Middle brands had a higher probability. Equivalently, there was approximately a 79.2% posterior probability that Lower-End brands had a higher probability of being classified as Outdated than Middle brands. These results suggest that Lower-End brands may have a greater probability of producing Outdated items, particularly when compared with Middle brands, although the uncertainty around these differences remains substantial.

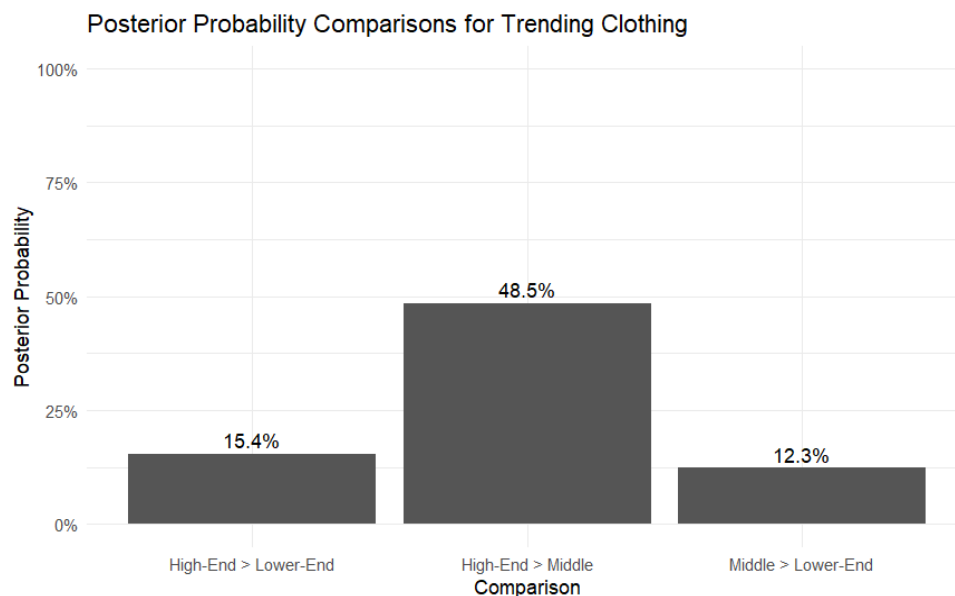


Figure 10: Bar chart of the posterior probability comparisons for the trending trend status

For the Trending category, the difference between High-End and Middle brands was -0.001, with a 95% credible interval of (-0.176, 0.188) and a posterior probability of 48.5% that High-End brands had a higher probability. The difference between High-End and Lower-End brands was -0.093, with a 95% credible interval of (-0.262, 0.091) and a posterior probability of 15.4% that High-End brands had a higher probability. This corresponds to an approximately 84.6% posterior probability that Lower-End brands had a higher probability of being classified as Trending than High-End brands. Likewise, the difference between Middle and Lower-End brands was -0.092, with a 95% credible interval of (-0.246, 0.065) and a posterior probability of 12.3% that Middle brands had a higher probability. Thus, there was approximately a 87.7% posterior probability that Lower-End brands had a higher probability of being classified as Trending than Middle brands. These results suggest that Lower-End brands may be more likely to contain Trending items than the other market segments, although the credible intervals indicate considerable uncertainty about the magnitude of these differences.

Overall, the strongest differences were observed in the Emerging category. The posterior probability that High-End brands had a greater probability of being Emerging than Lower-End brands was 93.7%, while the corresponding probability for Middle versus Lower-End brands was 95.6%. The results also suggested that Lower-End brands may have higher probabilities of producing both Outdated and Trending items, while the probability of being Classic appeared relatively similar across the three market segments. However, all of the 95% credible intervals contained zero, meaning that none of the comparisons provided a 95% credible interval that excluded the possibility of no difference. Therefore, the results should be interpreted as varying degrees of posterior evidence rather than definitive differences between market segments.

Limitations and Further Analysis

One limitation of this analysis is that the classification of brands into High-End, Middle, and Lower-End categories was based on general market positioning, price, reputation, and fashion positioning rather than a quantitative measure. Because individual products within a brand can vary substantially in price and style, some products may not fit perfectly within their assigned market segment. Another limitation is the relatively small and potentially uneven sample of clothing items across brands. Some brands contribute substantially more observations than others, which may cause certain brands to have a greater influence on the posterior distributions of their respective market segments. In addition, the Trend_Status variable is a categorical classification already provided by the dataset. As a result, the analysis evaluates differences in the assigned classifications rather than independently measuring the popularity or fashionability of each item. The analysis also did not account for other potentially important variables, such as price, material, style, gender, or customer ratings. These characteristics may be related to both a brand's market positioning and the trend status of its products, making them potentially important factors for future modeling.

Future analysis could examine price directly rather than relying solely on the High-End, Middle, and Lower-End groupings. A model could investigate whether the probability of an item being Classic, Emerging, Outdated, or Trending changes as price increases. A multinomial regression model could also incorporate multiple predictors simultaneously, such as price, material, style,

gender, and season. This would allow us to determine whether market positioning remains associated with trend status after accounting for other characteristics of the clothing items. Finally, future research could extend the multinomial–Dirichlet framework across individual seasons to examine whether the probabilities of Classic, Emerging, Outdated, and Trending items change over time within each market segment.

Conclusion

Overall, this analysis provided insight into how fashion trend classifications differ across brands and broader market segments. By grouping the brands into High-End, Middle, and Lower-End categories and modeling the four trend classifications using a multinomial distribution with a Dirichlet prior, we were able to estimate the probability that a clothing item within each market segment would be classified as Classic, Emerging, Outdated, or Trending. The posterior probabilities showed that Outdated had the highest estimated probability within each of the three market segments, while the probability of being Classic was relatively similar across groups.

The Monte Carlo comparisons provided further insight into the differences between market segments. The strongest evidence was found for the Emerging category, where there was a 93.7% posterior probability that High-End brands had a greater probability of being Emerging than Lower-End brands and a 95.6% posterior probability that Middle brands had a greater probability than Lower-End brands. The results also suggested that Lower-End brands may have higher probabilities of being both Outdated and Trending, although the uncertainty surrounding these differences was greater. Overall, the findings suggest that market positioning may be associated with how clothing items are distributed across different stages of the fashion cycle, particularly for Emerging, Trending, and Outdated products. However, the results should be interpreted cautiously because all of the 95% credible intervals included zero, indicating that the data do not provide highly certain differences between the groups.

Despite these limitations, the analysis demonstrates how Bayesian methods can be used to examine categorical fashion trends while accounting for uncertainty in estimated probabilities. Future research incorporating additional variables such as price, material, style, customer ratings, and season could provide a more complete understanding of the factors associated with fashion trend cycles.

AI Disclosure

AI was used in this research to assist with R coding for data manipulation, Bayesian multinomial–Dirichlet modeling, Monte Carlo simulations, and the interpretation and summarization of model results. AI was also used to help check the accuracy and functionality of the code and to clarify the interpretation of the statistical analyses. Additionally, AI was used to review grammar, spelling, and organization throughout the paper to improve its clarity, coherence, and overall flow. All analyses, results, and conclusions were reviewed and verified by the authors.

